

1.6a

Sensation Basic Concepts

Prosopagnosia :: face blindness

people with this can't even recognize their own face

underside right hemisphere, most have a functioning area that helps recognize a familiar human face, in $\frac{1}{7}$ seconds.

Nature's sensory gifts enable animals to obtain essential info

Human ears sensitive to sound frequencies that include human voices, esp. baby cries

Frog, which feed on flying insects, have cells that fire only from small, dart, moving objects

Male silk worm moths can detect 10^{-9} ounce of chemical sex attractant from female 1 mile away.

Processing Sensations and Perceptions

A person with face-blindness.

Upon looking at a friend, their sensation is normal

Their sensory receptors are detecting the same info as most others & nervous system transmits that info to the brain

Their perception :: process by which brain organizes and interprets sensory input, is almost normal

They may recognize people from hair, voice, etc, just not face.

Normally sensation & perception blend into one continuous process

As brain absorbs info bottom-up processing enables sensory systems to detect lines, angles, colours, etc. comprising the image.

with, top-down processing, you interpret what your senses detect

What do sensory systems fundamentally do?

All senses:

- receive sensory stimulation, often using specialized receptor cells
- transform stimulation → neural impulses
- deliver neural info to brain

Sensory systems do **transduction** :: converting energy to info.

Vision processes light energy
Hearing processes sound waves

Psychophysics studies relationship between **physical energy** & effects on **psychological experience**

What info is perceived?

We are constantly exposed to much info, like x-rays, very high frequency sounds, that is **not needed**, so we are **blind** to it.

Other animals **sense different info**, depending on needs

Absolute thresholds

We are very **sensitive to some kinds** of stimuli

The **absolute threshold** is the **edge of our awareness** of these faint stimuli.

To test AT for sound, varying tones sent into each ear, and recorded whether heard.

The test would find a point where you could detect stimuli 50% of the time :: AT

Detecting weak stimuli depend on experience, expectations motivation & alertness

Signal dependence theory predicts when we will detect weak signals. (measured 'hits' : 'false')

SD theorists look at individual differences for same stimuli & different for same person in different environment.

Stimuli u detect < 50% of time are subliminal

Subliminal sensations too weak to be perceived can still affect subconscious decisions.

Subliminal stimuli can prime us but can't persuade us to change behavior.

Difference thresholds.

We need to be able to detect small discrepancies in stimuli

Difference Threshold :: min Δ stimulus a person can detect 50% of the time; just noticeable difference

Detectable Difference $\propto$ size of stimuli

Weber's law: for any person to perceive difference, two stimuli must differ by a constant min percentage, not constant amount.

How do our senses adapt?

Sensory Adaptation :: continuous exposure to an unchanging stimulus lessens our awareness of it, as our nerve cells fire less frequently

Our eyes are constantly flitting from one spot to another, which is why objects don't disappear from views.

Using a device that moved an image when the eye moved, participants see fragments disappear & reappear.

Sensory adaptation can affect our perception of emotions. Focusing on a face with a certain emotion will temporarily make it "default", portraying others relative to this one.

Our sensory system is alert for novelty, repetition will lead it to free our mind for more important things.

We perceive the world not exactly as it is, but as it is useful for us to perceive it.

MCQ: 1c 2a 3c 4d 5b 6c